Contents

Hubble Tension Route Map - 2026 Edition

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Claim boundary

This is an educational route map. It does not claim to solve the Hubble tension, propose a new cosmological model, overturn Lambda-CDM, or provide expert-reviewed conclusions. It is a map for reading the existing literature.

Bilingual edition note

The Japanese and English editions share the same chapter structure and claim boundary, but they are parallel educational editions rather than strict word-for-word translations. Wording may differ to improve readability in each language, while the scientific scope remains aligned.

01. What is the Hubble tension?

Purpose of this chapter

This chapter explains What is the Hubble tension? as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> The Hubble tension is not merely a disagreement over one number; it is a consistency problem across multiple observational routes.

Background

The first mistake to avoid is treating the Hubble tension as a conflict between two isolated numbers. The number H0 appears simple, but each route to it carries a different observational chain, calibration structure, likelihood, and model assumption.

The issue is therefore not a vote between two measurements. It is a question of whether early-universe inference, local distance calibration, standard-ruler measurements, supernova distances, and structure-growth constraints can all live inside the same cosmological picture.

What the reader should learn

For a beginner, the safest entry point is to separate what is directly observed from what is inferred through a model. This prevents the topic from becoming a premature story about a solved mystery or a single hidden error.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- The issue cannot be understood by H0 values alone.
- Each observational route has its own assumptions and vulnerabilities.
- This handbook is a constraint map, not a proposed solution.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.
- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.

02. H0, distance, redshift, and expansion

Purpose of this chapter

This chapter explains H0, distance, redshift, and expansion as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> H0 is not a standalone measurement; it gains meaning through distance definitions, redshift, calibration, and cosmological assumptions.

Background

H0 is often introduced as the present expansion rate of the Universe, but reaching a value for H0 requires a chain of distance definitions, redshift interpretation, calibration, and model assumptions.

In the local route, a distance ladder connects nearby anchors to farther supernovae. In the CMB route, early-universe observables are translated into a present-day H0 through a cosmological model such as Lambda-CDM.

What the reader should learn

Without this distinction, direct distance calibration and model-dependent inference can be confused under the same symbol. The Hubble tension is clearer once distance, redshift, and expansion history are separated.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- H0 depends on distance scale and redshift interpretation.
- Local and CMB routes infer H0 differently.
- The same symbol does not imply the same measurement route.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.

03. Why Lambda-CDM remains the standard baseline

Purpose of this chapter

This chapter explains Why Lambda-CDM remains the standard baseline as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Lambda-CDM remains the baseline not because it is perfect, but because it connects many observations with a compact parameter set.

Background

Lambda-CDM functions as the standard model of cosmology because it has connected the CMB, BAO, supernova distances, large-scale structure, and other observations with a compact parameter set.

Being the baseline does not mean being free of tension. Hubble tension and S8 tension are diagnostic points where the success and limits of the baseline are tested.

What the reader should learn

A proposed extension must therefore do more than improve H0. It must preserve, or improve, the broader observational network that Lambda-CDM already organizes.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Lambda-CDM is a baseline, not an infallible theory.
- Alternatives must preserve existing successes, not only shift H0.
- Dissatisfaction with a baseline is not the same as a successful replacement.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

04. What a cosmological tension means

Purpose of this chapter

This chapter explains What a cosmological tension means as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> A cosmological tension is a diagnostic state depending on data, model assumptions, priors, likelihoods, and statistical evaluation.

Background

A tension is not simply a disagreement of opinions. It is a statistical diagnostic of how difficult it is to fit multiple datasets or inference routes within the same model.

The strength of a tension is not an absolute number. It changes with dataset choices, free parameters, priors, likelihood construction, and model comparison criteria.

What the reader should learn

Therefore, a sigma value should never be read alone. It must be tied to the data, assumptions, and statistical method that produced it.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- A tension is a statistical diagnostic.
- Its strength depends on data choices and priors.
- A sigma value is unsafe without its assumptions.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

05. CMB: early-universe inference

Purpose of this chapter

This chapter explains CMB: early-universe inference as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> The CMB does not directly measure H0; it infers H0 by translating early-universe features through a cosmological model.

Background

The CMB contains information from the early hot Universe. Temperature and polarization patterns are measured with high precision and constrain cosmic composition and initial conditions.

The H0 value inferred from the CMB is not a direct local measurement. It is obtained by translating angular scales, matter density, sound horizon physics, and other observables through a model such as Lambda-CDM.

What the reader should learn

Moving the CMB-inferred H0 therefore requires preserving early-universe physics, recombination, the sound horizon, matter density, and the detailed structure of the CMB peaks.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- CMB H0 is model-inferred.
- The CMB is a multi-feature constraint.
- Moving H0 alone can damage the CMB peak structure.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.

06. The distance ladder

Purpose of this chapter

This chapter explains The distance ladder as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> The distance ladder is a late-universe route that connects nearby distance anchors to calibrated Type Ia supernovae.

Background

The distance ladder begins with nearby anchors and extends the distance scale outward using standard candles such as Cepheids. It eventually calibrates Type Ia supernovae in the Hubble flow to infer a local H0.

Its strength is that it is relatively independent of early-universe assumptions. Its vulnerability is that each rung requires careful calibration and control of selection, crowding, metallicity, photometry, and supernova standardization.

What the reader should learn

The distance ladder should therefore be read as a chain of anchors, Cepheids, supernovae, and Hubble-flow objects, not as a single measurement.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- The distance ladder is a late-universe route.
- Each rung requires calibration and systematics checks.
- The chain matters as much as the final H0 value.

Related references

- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.

07. TRGB and JAGB as independent distance routes

Purpose of this chapter

This chapter explains TRGB and JAGB as independent distance routes as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> TRGB and JAGB provide independent distance indicators for testing whether a particular local-distance route hides systematics.

Background

If the distance ladder is read only through the Cepheid route, systematics tied to one stellar population or observational environment may be missed. Independent indicators such as TRGB and JAGB are therefore important.

TRGB uses the luminosity of the tip of the red giant branch, while JAGB uses properties of carbon-rich asymptotic giant branch stars in the near-infrared. Their different astrophysics make them useful cross-checks.

In the JWST era, CCHP-related work compares TRGB, JAGB, and Cepheid routes across nearby supernova host galaxies. This is educationally important because the local-distance question should be read as a comparison among routes, not as a verdict from one isolated number.

What the reader should learn

However, an independent indicator is not an automatic verdict. Calibration, sample selection, environmental dependence, and statistical treatment must still be compared carefully.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- TRGB and JAGB cross-check the local distance scale.
- They test systematics not visible in a single route.
- JWST-era comparisons require reading several local-distance indicators together.
- Independence matters, but does not by itself settle the problem.

Related references

- Freedman et al. The Carnegie-Chicago Hubble Program. VIII. An Independent Determination of the Hubble Constant Based on the Tip of the Red Giant Branch. Astrophysical Journal 882, 34 (2019). arXiv:1907.05922.
- Lee et al. The Chicago-Carnegie Hubble Program: The JWST J-region Asymptotic Giant Branch (JAGB) Extragalactic Distance Scale. arXiv:2408.03474.
- Freedman et al. Status Report on the Chicago-Carnegie Hubble Program (CCHP): Three Independent Astrophysical Determinations of the Hubble Constant Using the James Webb Space Telescope. arXiv:2408.06153.

08. BAO and the sound horizon

Purpose of this chapter

This chapter explains BAO and the sound horizon as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> BAO constrains distances relative to the sound horizon, making it a major failure boundary for many proposed H0 routes.

Background

BAO uses the imprint of early-universe acoustic oscillations in large-scale structure as a standard ruler. The characteristic scale in galaxy clustering provides distance information across redshift.

Its importance lies not in being a single direct H0 measurement, but in tying the sound horizon to the expansion history. Any model that changes the early sound horizon must remain compatible with BAO distances.

What the reader should learn

A proposal may appear to raise H0 successfully, yet fail when BAO is included. BAO is therefore a crucial failure boundary for proposed solutions.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- BAO constrains distances relative to the sound horizon.
- Early-universe modifications must pass BAO.
- BAO is a key failure boundary.

Related references

- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.
- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.

09. Type Ia supernovae and cosmic distances

Purpose of this chapter

This chapter explains Type Ia supernovae and cosmic distances as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Type Ia supernovae function both as calibrated distance-ladder objects and as relative-distance probes in cosmology.

Background

Type Ia supernovae are widely used because their luminosities can be standardized. In the local distance ladder, calibrated supernovae connect nearby distance anchors to H0.

Supernova samples can also be used as relative-distance probes, independent of absolute calibration. In that role, they constrain expansion history and dark-energy behavior.

What the reader should learn

Thus, the reader must distinguish whether supernovae are being used in an H0 calibration chain or as a relative-distance cosmological probe.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Type Ia supernovae have two roles.
- Absolute calibration and relative distances must be separated.
- Supernovae are central constraints on expansion history.

Related references

- Brout et al. The Pantheon+ Analysis: Cosmological Constraints. Astrophysical Journal 938, 110 (2022). arXiv:2202.04077.
- Scolnic et al. The Pantheon+ Analysis: The Full Data Set and Light-curve Release. Astrophysical Journal 938, 113 (2022). arXiv:2112.03863.

10. S8 and structure growth

Purpose of this chapter

This chapter explains S8 and structure growth as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> S8 is not a local H0 measurement; it tests whether the growth of structure is compatible with CMB-inferred cosmology.

Background

S8 combines matter density and the amplitude of structure. It is constrained by weak lensing, galaxy clustering, and related probes. It is not H0, but it is tied to model consistency.

Some models that raise H0 also alter structure growth, potentially worsening S8 agreement. A model can reduce one tension while creating another.

What the reader should learn

S8 therefore tests whether a proposal handles perturbations and growth of structure, not only background expansion.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- S8 is not an H0 measurement.
- Improving H0 can worsen S8.
- Background expansion and structure growth must be tested together.

Related references

- Asgari et al. KiDS-1000 Cosmology: Cosmic shear constraints and comparison between two point statistics. Astronomy & Astrophysics 645, A104 (2021). arXiv:2007.15633.
- Abbott et al. DES Y3 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing. Physical Review D 105, 023520 (2022). arXiv:2105.13549.

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

11. DESI and JWST-era updates

Purpose of this chapter

This chapter explains DESI and JWST-era updates as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> DESI and JWST sharpen different parts of the map, but neither alone determines the physical cause of the Hubble tension.

Background

DESI sharpens BAO measurements and the large-scale distance map. JWST can strengthen checks of the local distance ladder, especially crowding and photometric systematics.

These are both important, but they do not provide the same kind of information. DESI mainly constrains standard-ruler distances and expansion history; JWST tests aspects of local calibration.

JWST itself should not be read as a single route. Riess et al. use JWST Cepheid observations to test whether unrecognized crowding in HST Cepheid photometry could explain the tension. CCHP-related JWST work instead compares TRGB, JAGB, and Cepheid distance indicators to examine local-distance systematics from another angle.

What the reader should learn

Thus, DESI or JWST results should not be read as ending the whole problem. They update the map, but they do not automatically solve the full constraint network.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- DESI and JWST strengthen different constraints.
- Neither is a standalone final answer.
- JWST has several educationally distinct roles: Cepheid-systematics testing and independent-distance-indicator comparison.
- Recent observations update the map rather than close it.

Related references

- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.

- Riess et al. JWST Observations Reject Unrecognized Crowding of Cepheid Photometry as an Explanation for the Hubble Tension. arXiv:2401.04773.
- Freedman et al. Status Report on the Chicago-Carnegie Hubble Program (CCHP): Three Independent Astrophysical Determinations of the Hubble Constant Using the James Webb Space Telescope. arXiv:2408.06153.

12. Observational systematics

Purpose of this chapter

This chapter explains Observational systematics as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> A systematic explanation must be large enough, have the right direction, and remain compatible with other data.

Background

Observational systematics are always a serious possibility in the Hubble tension. Photometry, crowding, calibration, sample selection, and stellar environments all matter for local distance measurements.

But merely naming a systematic is not an explanation. The effect must be large enough, have the correct sign, and remain compatible with independent tests.

What the reader should learn

Systematics may look less dramatic than new physics, but they are essential. Before invoking new physics, the vulnerabilities of each observational route must be quantified.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Systematics must always be tested.
- Magnitude, direction, and independent consistency matter.
- Observational vulnerabilities must be quantified before new physics is claimed.

Related references

- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.

- Freedman et al. The Carnegie-Chicago Hubble Program. VIII. An Independent Determination of the Hubble Constant Based on the Tip of the Red Giant Branch. Astrophysical Journal 882, 34 (2019). arXiv:1907.05922.
- Riess et al. JWST Observations Reject Unrecognized Crowding of Cepheid Photometry as an Explanation for the Hubble Tension. arXiv:2401.04773.

13. Early Dark Energy

Purpose of this chapter

This chapter explains Early Dark Energy as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Early Dark Energy aims to reduce the sound horizon and raise H0, but it is tightly constrained by CMB and structure-growth consistency.

Background

Early Dark Energy introduces a temporary energy component in the early Universe, often around the pre-recombination era, to reduce the sound horizon and raise the CMB-inferred H0.

Its appeal is that it acts on the sound horizon, a central lever in the tension. But it must also preserve CMB peak structure, matter density, structure growth, and S8 behavior.

What the reader should learn

Early Dark Energy is therefore an important proposal route, not a settled solution. Its viability depends on parameter space, dataset combinations, and model comparison.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- EDE targets the sound horizon.
- CMB and S8 constraints are severe.
- A promising idea is not the same as an established solution.

Related references

- Poulin et al. Early Dark Energy Can Resolve The Hubble Tension. Physical Review Letters 122, 221301 (2019). arXiv:1811.04083.
- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.
- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.

14. Dynamic dark energy

Purpose of this chapter

This chapter explains Dynamic dark energy as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Dynamic dark energy changes late-time expansion history, but hints in expansion history and resolution of H0 tension are different claims.

Background

Dynamic dark energy considers the possibility that dark energy evolves with time. It affects late-time expansion history and is especially relevant to supernova and BAO interpretation.

However, evidence for flexibility in expansion history is not the same as resolving the Hubble tension. H0, BAO, supernovae, CMB inference, and growth of structure must all remain compatible.

What the reader should learn

The reader should distinguish expansion-history flexibility, dataset dependence, and apparent improvement from extra parameters.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Dynamic dark energy affects late expansion.
- Expansion-history hints do not equal H0 resolution.
- Parameter flexibility must be read cautiously.

Related references

- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.
- Brout et al. The Pantheon+ Analysis: Cosmological Constraints. Astrophysical Journal 938, 110 (2022). arXiv:2202.04077.

15. Modified gravity

Purpose of this chapter

This chapter explains Modified gravity as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Modified gravity affects not only background expansion but also perturbations, lensing, and gravitational-wave constraints.

Background

Modified gravity refers to proposals that extend or alter the cosmological behavior of general relativity. Changing background expansion can affect H0, but the evaluation does not stop there.

Changing gravity also affects growth of structure, lensing, galaxy clustering, and gravitational-wave propagation. In some models, gravitational-wave speed and effective gravitational strength are decisive constraints.

After GW170817/GRB 170817A, comparisons between gravitational-wave and electromagnetic arrival times placed strong constraints on classes of modified-gravity models that shift the gravitational-wave speed away from the speed of light. A modified-gravity route must therefore be read against multi-messenger constraints, not only against H0.

What the reader should learn

Thus, modified gravity routes must be tested across background, perturbations, and multi-messenger constraints.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Modified gravity is not only background expansion.
- Perturbations, lensing, and gravitational waves matter.
- GW170817/GRB 170817A is a key example of a strong gravitational-wave speed constraint.
- H0 alone cannot validate a gravity modification.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.
- Abbott et al. A gravitational-wave standard siren measurement of the Hubble constant. Nature 551, 85-88 (2017). arXiv:1710.05835.
- Abbott et al. Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A. Astrophysical Journal Letters 848, L13 (2017). arXiv:1710.05834.

16. Dark radiation and N_eff

Purpose of this chapter

This chapter explains Dark radiation and N_eff as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Changing N_eff can affect early expansion and the sound horizon, but it is strongly constrained by the CMB and primordial physics.

Background

N_eff represents the effective number of relativistic species in the early Universe. Extra light particles or dark radiation could change this value.

Increasing N_eff can change the early expansion rate and the sound horizon, thereby affecting inferred H0. But CMB damping, primordial nucleosynthesis, matter-radiation equality, and other constraints also respond.

What the reader should learn

This route is conceptually simple, but N_eff cannot be adjusted freely. It must pass several early-universe constraints simultaneously.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- N_eff tracks early relativistic energy density.
- It can affect H0, but constraints are strong.
- CMB and primordial physics must be considered together.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

17. Dark-sector interactions

Purpose of this chapter

This chapter explains Dark-sector interactions as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Dark-sector interactions must be evaluated through background expansion, perturbation stability, and structure growth.

Background

If dark matter and dark energy interact, energy transfer can modify expansion history and structure growth. Such models can affect both H0 and S8.

But interacting dark-sector models can introduce instabilities and parameter degeneracies. A background-level improvement may fail at the perturbation level or conflict with structure formation.

What the reader should learn

This route therefore requires simultaneous checks of H0, S8, perturbation stability, and the full observational dataset.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Dark-sector interactions can affect both H0 and S8.
- Perturbation stability is essential.
- Parameter degeneracy must be handled carefully.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

18. Recombination and early-universe physics

Purpose of this chapter

This chapter explains Recombination and early-universe physics as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Changing recombination can strongly affect CMB inference, but it is tightly constrained by CMB peak and damping structure.

Background

Recombination is the process in which free electrons and nuclei combine into neutral atoms, and it is deeply connected to the formation of the CMB. Beginner explanations often compress recombination, photon decoupling, and the last-scattering surface into one story, but these terms should not be treated as exact synonyms.

As recombination reduces the free-electron abundance, photons scatter less frequently and eventually travel almost freely through the Universe. This photon decoupling and last-scattering structure provides the information observed as the CMB. Modifying this history can change the sound horizon and angular scales inferred from the CMB.

But recombination also shapes the detailed CMB spectrum: peak positions, relative heights, damping tail, and polarization. These features must remain consistent.

What the reader should learn

Recombination-based proposals are powerful but tightly constrained. The question is not only whether H0 moves, but whether the full CMB structure survives.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Recombination, photon decoupling, and last scattering are closely related, but they should not be collapsed into exact synonyms.
- Recombination directly shapes the CMB.
- Peak and damping structures constrain modifications.
- The full CMB must be preserved, not only H0.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

19. Local structure and cosmic variance

Purpose of this chapter

This chapter explains Local structure and cosmic variance as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Local structure can affect local H0 measurements, but the effect must be large enough and in the right direction.

Background

Our local Universe need not be perfectly average. Local over-density or under-density can affect peculiar velocities and the local distance-redshift relation.

To explain the Hubble tension, however, the effect must be large enough, have the right sign, and remain consistent with galaxy distributions and velocity-field observations.

What the reader should learn

Local-structure explanations are intuitive, but quantitatively constrained. They must be tested rather than assumed.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Local structure can affect velocity fields.
- Magnitude and sign matter.
- Local-bias explanations require quantitative testing.

Related references

- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.
- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

20. Statistics, data integration, and priors

Purpose of this chapter

This chapter explains Statistics, data integration, and priors as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Fit improvement, model comparison, priors, and dataset selection must be evaluated separately.

Background

Cosmology combines multiple datasets through likelihoods, priors, covariances, correlations, and model parameters. Each of these choices can affect the result.

Adding parameters can improve the fit to a dataset, but that does not automatically provide a better physical explanation. Model comparison must account for both improvement and added freedom.

What the reader should learn

Beginners should look beyond best-fit values and ask which priors, datasets, and comparison criteria were used.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Statistical setup affects results.
- Fit improvement is not automatically explanation.
- Priors and model comparison must be checked.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

21. What breaks when H0 is raised?

Purpose of this chapter

This chapter explains What breaks when H0 is raised? as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> The difficulty is not simply raising H0; it is preserving CMB, BAO, supernova, and S8 consistency afterward.

Background

Many levers appear capable of raising H0: changing early-universe physics, reducing the sound horizon, modifying late expansion, or altering gravity.

Every lever has side effects. Changing the sound horizon affects BAO. Changing expansion history affects supernova distances and CMB connection. Changing perturbations affects S8 and large-scale structure.

What the reader should learn

Thus, when a model raises H0, the first question is what it sacrifices. A promising model must be read through its failure boundaries.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- There are many ways to raise H0.
- Each route has side effects.
- Failure boundaries are essential for judging proposals.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.
- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.
- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

22. CMB constraints as a multi-feature test

Purpose of this chapter

This chapter explains CMB constraints as a multi-feature test as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> The CMB is not a single H0 number but a high-dimensional constraint involving peaks, damping, polarization, and matter density.

Background

CMB constraints are often summarized by an H0 value, but the data include peak positions, peak heights, damping tail, polarization, lensing, and matter density information.

If a model moves CMB-inferred H0, it must still explain this full feature set. Matching one number is not enough if the spectral shape is damaged.

What the reader should learn

The CMB should therefore be read as a multi-feature test, not merely as the source of a low H0 value.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- The CMB is high-dimensional.

- Matching H0 while breaking spectra is failure.
- Peaks, damping, and polarization must be preserved.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Poulin et al. Early Dark Energy Can Resolve The Hubble Tension. Physical Review Letters 122, 221301 (2019). arXiv:1811.04083.

23. BAO constraints as a failure boundary

Purpose of this chapter

This chapter explains BAO constraints as a failure boundary as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Any route that changes the sound horizon must pass BAO consistency across redshift.

Background

BAO is the trace of an early-universe standard ruler in late-time large-scale structure. It links early-universe modifications to the late expansion history.

Models that reduce the sound horizon to address H0 must also explain BAO distance ratios across redshift. If those distances fail, the apparent H0 improvement is not sufficient.

What the reader should learn

BAO is therefore less a dramatic protagonist than a gate that proposed solutions must pass.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- BAO links early and late cosmology.
- Sound-horizon changes must fit BAO distances.
- H0 improvement cannot compensate for BAO failure.

Related references

- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.
- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.

24. The H0-S8 tradeoff

Purpose of this chapter

This chapter explains The H0-S8 tradeoff as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> H0 and S8 are distinct quantities, but they can trade off through model-dependent background and perturbation behavior.

Background

Models that improve H0 often alter expansion history or matter density. Those changes can propagate into S8, the amplitude of structure growth.

S8 is constrained by weak lensing and galaxy clustering. Whether late-time measurements agree with CMB-inferred expectations depends on the perturbation behavior of the model.

What the reader should learn

H0 and S8 should therefore be examined together inside the same model, rather than treated as independent scorecards.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- H0 improvement can affect S8.
- S8 tests growth consistency.
- Both must be evaluated in the same model.

Related references

- Asgari et al. KiDS-1000 Cosmology: Cosmic shear constraints and comparison between two point statistics. Astronomy & Astrophysics 645, A104 (2021). arXiv:2007.15633.
- Abbott et al. DES Y3 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing. Physical Review D 105, 023520 (2022). arXiv:2105.13549.
- Poulin et al. Early Dark Energy Can Resolve The Hubble Tension. Physical Review Letters 122, 221301 (2019). arXiv:1811.04083.

25. Parameter addition and overfitting

Purpose of this chapter

This chapter explains Parameter addition and overfitting as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Adding parameters can improve a fit, but it does not necessarily provide a better physical explanation.

Background

Adding a new parameter increases flexibility. It can make it easier to reduce a specific mismatch in a dataset.

But a better fit caused only by extra freedom is not necessarily a better physical explanation. The question is whether the data require the new component or whether the model is overfitting.

What the reader should learn

Information criteria, Bayes factors, prior dependence, and performance on external datasets are therefore important.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Flexibility can improve fits.
- Improvement is not necessarily explanation.
- External tests and model comparison are needed.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

26. Single-cause and composite explanations

Purpose of this chapter

This chapter explains Single-cause and composite explanations as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Composite explanations can be realistic, but each component needs independent support and testability.

Background

The Hubble tension may not have a single cause. Several smaller effects could combine: local systematics, early-universe physics, and statistical choices.

Composite explanations can be realistic, but they are not free. Each component must be independently supported, quantitatively sized, and directionally appropriate.

What the reader should learn

When reading composite explanations, the key question is whether explanatory power increased or whether the model merely gained escape routes.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Composite causes are possible.
- Each component needs independent support.
- Explanatory power must be distinguished from escape routes.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

27. How to read review papers

Purpose of this chapter

This chapter explains How to read review papers as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Review papers provide the map, but exact numerical and evidential claims must be checked in primary literature.

Background

Review papers are useful because they organize a large literature and identify major routes and constraints. For beginners, they provide a map before entering primary papers.

But a review is not the source of every exact number or figure. Important values, datasets, and statistical methods should be traced back to primary literature.

What the reader should learn

When reading reviews, the reader should note classification choices, emphasis, and publication date. Cosmology changes quickly enough that older reviews cannot be the whole current picture.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Reviews are useful maps.
- Numbers and evidence require primary sources.
- Publication date and author framing matter.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

28. Quantities and symbols

Purpose of this chapter

This chapter explains Quantities and symbols as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> H0, H(z), h, r_d, sigma8, and S8 are not interchangeable; each has a distinct physical meaning.

Background

Cosmology uses many similar symbols. H0 is the present expansion rate, H(z) is the expansion rate at redshift z, and h is H0 divided by 100 km/s/Mpc.

r_d is the sound horizon at the drag epoch, sigma8 is the amplitude of matter fluctuations on 8 h^-1 Mpc scales, and S8 combines sigma8 with matter density. These quantities are related but not identical.

What the reader should learn

Confusing symbols leads to misreading the Hubble tension. The reader should ask which quantity is observed, which is inferred, and which is derived.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Similar symbols can mean different things.
- r_d and S8 are not H0.
- Observed, inferred, and derived quantities must be separated.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.
- Asgari et al. KiDS-1000 Cosmology: Cosmic shear constraints and comparison between two point statistics. Astronomy & Astrophysics 645, A104 (2021). arXiv:2007.15633.

29. Common beginner misconceptions

Purpose of this chapter

This chapter explains Common beginner misconceptions as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> Many beginner misconceptions arise by compressing real technical issues into overly simple conclusions.

Background

Common misconceptions include saying that the CMB directly measures present-day H0, that the distance ladder alone proves new physics, that JWST ended the tension, or that DESI by itself provides the final answer.

Each misconception contains a real technical issue. The CMB is precise, the distance ladder matters, and JWST and DESI provide important constraints. The error is compressing them into a single definitive conclusion.

What the reader should learn

Beginners should respond to strong claims by unpacking the observational route, model assumptions, and constraint conditions behind them.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Misconceptions often contain real issues.
- Over-compression creates false conclusions.
- Strong claims should be unpacked.

Related references

- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.
- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.
- Riess et al. JWST Observations Reject Unrecognized Crowding of Cepheid Photometry as an Explanation for the Hubble Tension. arXiv:2401.04773.

30. Using AI to learn the Hubble tension

Purpose of this chapter

This chapter explains Using AI to learn the Hubble tension as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> AI can help organize learning, but AI output is neither observational evidence nor expert review.

Background

AI can help organize terminology, chapter structure, reading order, and conceptual relations. For beginners, it can make a broad field easier to enter.

However, AI explanation is not scientific evidence. AI can produce plausible but wrong statements and can overgeneralize what the literature supports.

What the reader should learn

AI should be used as a reading assistant, not as an authority. Claims must be returned to primary literature, reviews, observations, and reproducible checks.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- AI is useful as a learning aid.
- AI output is not evidence.
- Claims must return to literature and verification.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.

31. Next reading map

Purpose of this chapter

This chapter explains Next reading map as one entry point into the current Hubble tension landscape. The handbook does not propose a new model; it provides a bounded reading map for existing literature and observational probes.

The central thesis of this chapter is:

> The next learning step is to read the relation between observational probes and failure boundaries before choosing any favored route.

Background

After this handbook, a safe next step is to read a broad review, then move to primary papers on the CMB, distance ladder, BAO, supernovae, and S8.

The reader should not choose a favored solution first. It is better to learn which observations constrain which routes and where proposals tend to fail.

What the reader should learn

The goal is not to consume the Hubble tension as a grand story, but to learn how observations and theory are connected one constraint at a time.

This perspective is necessary for reading the Hubble tension as a network of observational routes and failure boundaries rather than as a simple question of moving H0 up or down.

Reading cautions

The reader should not use a single dataset, single paper, or isolated parameter value to make a definitive claim. The concept in this chapter must be read together with observational routes, model assumptions, and statistical treatment.

Key points

- Move from reviews to primary papers.
- Learn failure boundaries before favored solutions.
- Read the observation-theory connection carefully.

Related references

- Di Valentino et al. In the Realm of the Hubble tension - a Review of Solutions. Classical and Quantum Gravity 38, 153001 (2021). arXiv:2103.01183.
- Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astronomy & Astrophysics 641, A6 (2020). arXiv:1807.06209.
- Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. Astrophysical Journal Letters 934, L7 (2022). arXiv:2112.04510.
- DESI Collaboration. DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints (2025). arXiv:2503.14738.
- Poulin et al. Early Dark Energy Can Resolve The Hubble Tension. Physical Review Letters 122, 221301 (2019). arXiv:1811.04083.